AI.205

Reinforcement Learni

COURSE OVERVIEW

Learning from interaction: agents, rewards, policies, and value-based methods for sequential decision making.



ILLUSTRATIVE HERO IMAGERY

REINFORCEMENT LEARNING

1

SESSION

 60 MINUTES

ORIENTATION: LEARNING BY DOING

KEY TOPICS

- RL vs. supervised learning
- Agent–environment loop
- Rewards and returns
- Applications in robotics

TEACHING FOCUS

Learn optimal behavior from trial and error.

▲ **KEY TAKEAWAY** Reinforcement learning learns from consequences.

REINFORCEMENT LEARNING

2

SESSION

60 MINUTES

MARKOV DECISION PROCESSES

KEY TOPICS

- States, actions, transitions
- Rewards and discounting
- Policies and value functions
- Bellman equation

TEACHING FOCUS

Model sequential decisions formally.

KEY TAKEAWAY MDPs are the mathematical heart of RL.

REINFORCEMENT LEARNING

3

SESSION

60 MINUTES

DYNAMIC PROGRAMMING

KEY TOPICS

- Policy evaluation and improvement
- Policy iteration
- Value iteration
- Model-based planning

TEACHING FOCUS

Solve MDPs when the model is known.

KEY TAKEAWAY Dynamic programming computes optimal policies exac

REINFORCEMENT LEARNING

4

SESSION

60 MINUTES

MODEL-FREE RL

KEY TOPICS

- Monte Carlo methods
- Temporal difference learning
- Q-learning
- SARSA

TEACHING FOCUS

Learn without a model of the world.

KEY TAKEAWAY Model-free agents learn from experience alone.

REINFORCEMENT LEARNING

5

SESSION

⏱ 60 MINUTES

DEEP RL

KEY TOPICS

- Deep Q-networks (DQN)
- Policy gradients
- Actor–critic
- RL in continuous control

TEACHING FOCUS

Scale RL to high-dimensional robot tasks.

▲ **KEY TAKEAWAY** Deep RL lets robots learn complex skills.

6

SESSION

60 MINUTES

RL FOR ROBOTS & REVIEW

KEY TOPICS

- Sim-to-real transfer
- Safe exploration
- Course recap and assessment
- Toward robot learning (AI.403)

TEACHING FOCUS

Deploy learned policies on real robots.

KEY TAKEAWAY Simulation trains; reality tests.